Python Engineering - Class 04 Theory Guide

Functions and Decomposition

This guide explains the programming, mathematical and/or engineering theory behind the exercise. The goal is to understand the model before memorizing syntax.

1. Decomposition

Functions split a large problem into smaller units that can be understood, reused and validated independently.

Relationship / equation

```
problem = validation + model + integration + analysis
```

Connection to code

EX12 separates these responsibilities rather than mixing everything in one loop.

2. Inputs and outputs

A useful engineering function has a clear contract: what it receives, what assumptions apply, and what it returns.

Relationship / equation

```
f(inputs) -> outputs
```

Connection to code

Avoid hidden dependencies on global variables when the value naturally belongs in the function input.

3. Scope

Local variables protect intermediate calculations from accidental interactions with unrelated code.

Relationship / equation

```
local vs global namespace
```

Connection to code

A local variable can reuse a name without automatically changing a variable outside the function.

4. Return values

Printing is not the same as returning. A returned value can be tested, stored, plotted or passed to another function.

Relationship / equation

```
return data; print presentation
```

Connection to code

This distinction becomes essential when modules and tests are introduced later.

5. Functional validation

Small functions are easier to validate with known inputs and outputs.

Relationship / equation

```
known input -> function -> expected result
```

Connection to code

EX09-EX12 prepare the course for modular architecture and automated testing.

Manual validation method

For any exercise in this class, use the following protocol before trusting the output:

- 1. Write input values with units or types.
- 2. Predict sign, scale, type and shape of the result.
- 3. Manually calculate a small case when possible.
- 4. Run the code and compare against the prediction.
- 5. Change one input in a controlled way and explain the effect.

Connection to the rest of the course

Functions become modules, tests and reusable scientific libraries in later weeks.

Final summary

```
MODEL / CONCEPT
      |
      v
ALGORITHM
      |
      v
PYTHON IMPLEMENTATION
      |
      v
REFERENCE CHECK
      |
      v
ENGINEERING INTERPRETATION
```

> Syntax is the tool. The transferable skill is being able to model, structure, validate and interpret.